

QUALCOMM AI HUB HACKATHON | SNAPDRAGON® X NPU ACCELERATION

Snapdragon® Document Assistant

Offline On-Device Document Intelligence, Plain-Language Simplification & Audio Accessibility for Indian Languages

100% On-Device NPU

TrOCR & Qwen on Qualcomm Hexagon HTP v73

Fidelity Safeguard

Deterministic entity verification preventing LLM

True Offline Privacy

Zero cloud API calls, zero telemetry, memory-safe wipe

The Problem: The Literacy & Language Accessibility Barrier

Complex Legal & Medical Jargon

Official government notices, hospital invoices

Catastrophic Cost of Mistakes

Misinterpreting a deadline or payment amount

The Cloud & Hallucination Dilemma

Uploading sensitive medical bills to cloud storage

The Solution: 4-Stage On-Device Intelligence Pipeline

STAGE 1: OCR

TrOCR on NPU

Qualcomm QNN EP offload on Hexagon

STAGE 2: SIMPLIFY

Plain-Language LLM

Rewrites dense jargon to 6th-8th grades

STAGE 3: TRANSLATE

Indic Translation

Anchor-Preserved Translation interface

STAGE 4: SPEECH

Neural TTS

On-device Piper ONNX text-to-speech with spoken

Key Differentiator: Deterministic Cross-Stage Fidelity Safeguard



The Threat: LLM Hallucinations

- LLMs routinely drop dollar signs or digits (\$450 -> \$50).
- Polarity inversion: 'Claim Approved' translated as 'अस्वीकृत' (Rejected)
- Date shifts: Due dates altered from Oct 15 to Nov 15.
- Unacceptable in medical statements and legal summonses.



Our Solution: Deterministic Entity Verification

- Exact entity auditing across every stage (Dates, Amounts, IDs, Status).
- Cross-language regex auditing across English, Hindi, Tamil, Bengali.
- Spoken Hindi Warning prepended to speech if mismatch detected:
- 'चेतावनी: इस दस्तावेज़ में जानकारी की पुष्टि नहीं हो सकी है। मूल दस्तावेज़ की जाँच करें।'
- Strict Safety Gate option: Suppresses audio generation completely.

Qualcomm AI Hub & Hexagon NPU Acceleration

TrOCR Vision Encoder / Decoder

100% NPU Layer Placement

Compiled via Qualcomm AI Hub for Snapdragon

Genie LLM Runtime

Qwen3-1.7B w4a16

Quantized 4-bit weights and 16-bit activations

Sub-Millisecond Routing

Anchor-Preserved Lexicon

Structured bills and notices bypass generation overhead

Security, Privacy & Robustness Architecture

Zero Network Dependency

100% offline runtime verified under physical network disconnect. Zero runtime

Memory-Safe Data Retention

Document scans and extracted text exist strictly in volatile RAM. No unmanaged temp o

Hardened Against Attacks

Sanitized SAPI5 shell injection via stdin piping. Sanitized exception traces to p

Dependency Vulnerability Free

Audited via Google OSV / PyPI database: Transformers (0 CVEs), ONNX Runtime (0 CVEs)

Snapdragon Document Assistant: Ready for Real-World Impact

- ✓ **Production Packaging:** Compiled standalone Windows binary (SnapdragonDocAssistant.exe) with bundled offline models and accessible
- ✓ **Empowering Citizens:** Enables vulnerable users to independently understand complex medical and legal notices with complete privacy.
- ✓ **Proven Qualcomm Hardware Leverage:** Built specifically to demonstrate the speed, thermal efficiency, and intelligence of Snapdragon®
- ✓ **Open Source & Verifiable:** Complete source code, regression suites, offline benchmarks, and demo files available on GitHub: <https://github.com/Qualcomm-Innovation-Center/SnapdragonDocAssistant>